

Deva Anand

(413) 315-1715 | devaanand@umass.edu | devaanand.com | linkedin.com/in/devaaa | github.com/Deva-1903

Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Performance Engineering, Systems for Data Science, Optimization in Computer Science, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, Java, C++, JavaScript, TypeScript, SQL, Bash, HTML/CSS

Backend & APIs

Node.js, Express, FastAPI, Fastify, REST API design, client-server protocols, async services, integration testing, authentication, rate limiting

Data & Systems

Apache Spark / PySpark, Databricks, distributed data processing, ETL, Spark UI profiling, Linux performance analysis

Cloud & DevOps

AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD, Unix/Linux

Databases & Web

MongoDB, MySQL, PostgreSQL, Redis, SQLite, React, Electron, Webpack

Experience

Wysa

July 2023 – July 2025

Backend Engineer

Bengaluru, India

- Engineered a **multi-tenant NHS eTriage backend** for **8+ UK clients** (Node.js, MongoDB) that serves triage data to user-facing clinical applications, processing **10,000+ monthly submissions** through REST APIs and aggregation pipelines integrated with Mayden's iaptus platform, contributing to **\$340K+** in revenue.
- Architected a **full-stack data annotation platform (React, Node.js, MongoDB)** integrating an intuitive UI with secure role-based REST APIs and data-processing workflows, replacing spreadsheet processes and cutting manual operations by **100+ hours/month** while improving labeling throughput by **60%**.
- Owned reliability and security across the eTriage stack: debugged production logs during feature releases and client integrations, shipped fixes for tenant-specific incidents, hardened authentication and rate limiting, and remediated **20+ Vulnerability Assessment and Penetration Testing (VAPT) findings** protecting sensitive clinical data.

Cario Growth Services Pvt. Ltd.

November 2022 – May 2023

Backend Developer Intern

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js and Fastify backed by **PostgreSQL**, applying client-server API design with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated the core codebase from **JavaScript to TypeScript** across multiple repositories, strengthening object-oriented structure, type safety, and maintainability, and prototyped LLM-powered context-aware replies in a chat product.

Projects

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Developed a multi-stage agentic web-discovery service that plans typed retrieval, reranks pages, classifies evidence regimes, extracts entities with deterministic parsers before LLM fallback, verifies outputs, and returns cell-level provenance for every result.
- Hardened the service with **150+ automated tests**, provider routing and fallback across LLM backends, and pipeline instrumentation for predictable, debuggable behavior under changing inputs.

Spark ETL Performance Optimization - NYC TLC Dataset

github.com/Deva-1903/Spark-ETL-Optimization

Apache Spark / PySpark, Databricks, SQL, Python

- Optimized a distributed Spark ETL pipeline on the NYC TLC dataset using broadcast-join tuning, partitioning, caching, adaptive settings, and column pruning to reduce shuffle volume and wall-clock runtime across data-processing stages.
- Profiled stage-level timings and shuffle/skew through the Spark UI to localize bottlenecks via critical analysis, documenting a repeatable optimization workflow for downstream analytics jobs.

Sonare - Offline Sign ↔ Speech Cross-Platform Desktop App

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

- Built a **cross-platform desktop application** combining camera, microphone, UI, and backend services into a single on-device experience using MediaPipe hand tracking, gesture classification, whisper.cpp Automatic Speech Recognition (ASR), and local Text-To-Speech (TTS), designed and shipped under hackathon time pressure.
- Engineered a **FastAPI backend** and **Electron/React frontend** to run fully offline with no external network dependencies; designed async APIs and queues to keep the UI responsive while handling local video and audio streams.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.